



Some thesis title

by

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*Dissertation presented for the degree of Electrical and
Electronic Engineering in the Faculty of Engineering at
Stellenbosch University*

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2026

Someone.

ACKNOWLEDGEMENTS

* Supervisor X.

* Funding Y.

“ *A very long quote*
from someone very important. ”

– Mr M

Thank you to Leanne for helping to prepare this template.

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ABSTRACT

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NOMENCLATURE

ACRONYMS AND ABBREVIATIONS

MFT My Funky Thesis

DEFINITIONS

S

STUDENT is an entity needing a thesis to transcend the state of being a student.

INTRODUCTION

1.1 BACKGROUND

Given data from many hours of recordings which contains whale noises at undetermined times is difficult to analyse where these whale sounds occur at a glance. This is of interest as researchers can use this information to determine the migration patterns of whales and other behaviours.

1.2 PROBLEM STATEMENT

Determine whether using Dynamic Time Warping (DTW) in the frequency domain, thus using vector features, improves the ability of DTW to detect and classify whale calls compared to the time-domain approach of Dynamic Time Warping, thus using scalar features. The methods used will be Short Time Fourier Transform (STFT), Mel-frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC).

1.3 OBJECTIVES

The DTW method in the time domain will first be implemented. The scalar features will be used from the data as it is given in the file.

The DTW method in the frequency domain will then be implemented using three different extraction methods to ensure that the comparison between scalar and vector features does not depend on how the vector features are chosen. The methods used will be Short Time Fourier Transform (STFT), Mel-frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC).

The four models will then be compared to determine which method is best for whale detection and to determine whether the frequency domain approach is better and if so,

WELCOME TO ENGINEERING

Figure 1.1: Your figure caption.

how significant the improvements are.

1.4 THESIS OVERVIEW

Write a good thesis and follow the instructions at [1]. The pukka $\text{\LaTeX}$ template for Stellenbosch Theses is available at [2], this one is just for dummies.

My funky thesis (MFT) is as good as MFT can be, as Table 1.1 and Figure 1.1 shows. Have a look at my explanation in Section 2.1 on page 3, which is in Chapter 2.

Table 1.1: Your table caption

	Centered number [Units]	R-aligned number	Width-set col
Description	100	140	24601

LITERATURE

2.1 RELATED WORK

SUMMARY AND CONCLUSIONS

REFERENCES

- [1] Stellenbosch University, “Formatting your thesis,” 2024, <http://bit.ly/SUThesisFormat>.
- [2] D. Els, “Stellenbosch university thesis bundle,” 2024, <https://ctan.org/pkg/stellenbosch-2>.

APPENDIX A

APPENDIX TITLE GOES HERE
